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DETAIL SPECIFICATION

LEATHER, SHEEPSKIN, CHROME TANNED

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for chrome tanned sheepskin leather.

1.2 Classification. The leather is available in the following types, classes and subclasses as specified in 6.2.

1.2.1 Types.

Type A - Hair sheep
Type B - Domestic

1.2.2 Classes.

Class 1 - Mildew resistant treated
Class 2 - Not mildew resistant treated

1.2.3 Subclasses.

Subclass A - Perspiration resistant treated
Subclass B - Not perspiration resistant treated

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

AMSC N/A FSC 8330

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide and Rodenticide (FIFRA)
(40 CFR part 162 State Registration of Pesticides Products)

Federal Insecticide, Fungicide and Rodenticide (FIFRA)
as amended by the Food Quality Protection Act of 1996 and the Pesticide Registration Improvement Act of 2003

(Copies of these documents are available online at https://www.epa.gov/pesticides.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

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ASTM INTERNATIONAL

ASTM D2096 - Standard Test Method for Colorfastness and Transfer of Color in the Washing of Leather
ASTM D2211 - Standard Test Method for Elongation of Leather
ASTM D2322 - Standard Test Method for Resistance of Shoe Upper Leather to Artificial Perspiration
ASTM D2617 - Standard Test Method for Total Ash in Leather
ASTM D2807 - Standard Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)
ASTM D2810 - Standard Test Method for pH of Leather
ASTM D2813 - Standard Practice for Sampling Leather for Physical and Chemical Tests
ASTM D2821 - Standard Test Method for Measuring the Relative Stiffness of Leather by Means of a Torsional Wire Apparatus
ASTM D3495 - Standard Test Method for Hexane Extraction of Leather
ASTM D4705 - Standard Test Method for Stitch Tear Strength of Leather, Double Hole
ASTM D5053 - Standard Test Method for Colorfastness of Crocking of Leather
ASTM D6012 - Standard Test Method for Determination of Resistance of Leather to (Bleeding), Color Stain Transfer
ASTM D6076 - Standard Test Method for Shrinkage Temperature of Leather

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 13365-1 - Leather – Chemical Determination of the Preservative (TCMTB, PCMC, OPP, OIT) Content in Leather by Liquid Chromatography Part 1 – Acetonitrile Extraction Method

(Copies of this document are available online at https://www.iso.org.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Materials. The strap leather shall be full grain, chrome tanned sheepskin. The flesh side of the leather shall be smooth and free of coarse loose fiber.

3.3.1 Type A, hair sheep. Type A leather shall be made from skins of the hair sheep from Brazil, Nigeria, Sudan, Abyssinia, Djeddah or other origins. Wool or coarse wool sheepskins, such as Iranian type sheepskins, are not acceptable.

3.3.2 Type B, domestic. Type B leather shall be made from the skins of the domestic sheep.

3.3.3 Subclass A. When subclass A is specified, glutaraldehyde shall be incorporated into the leather.

3.4 Tannage. The leather shall be chrome tanned.

3.5 Color. The color of the leather, the method of application (drum, spray or brush) and the degree of dye penetration shall be as specified (see 6.2 and 6.3). When the spray or brush method is required, the surface of the leather to be dyed shall be as specified (see 6.2). Visual examination of the color and appearance shall be as specified in 4.6.1.

3.6 Finish. Unless otherwise specified (see 6.2), a light application of finish and dye containing only sufficient pigment as to assist in making the color uniform shall be applied to the grain surface. The finish shall not shrink or change the appearance of the natural grain surface, mar the leather, or camouflage any defect.

3.7 Physical requirements. The finished leather shall conform to the physical and chemical requirements listed below in Table I, when tested as specified in 4.5.

TABLE I. Physical and chemical requirements.

Characteristic	Requirement
Thickness (ounces)	1.25 - 2.50 1/
Stiffness (max.) 2/	60 degrees
Colorfastness to crocking (min.): Dry Wet	 3.0 2.0
Staining (min.)	4.0
Stitch tearing strength, pounds (min.) 2/ Type A Type B	 15 10
Elongation, percent (min.) 2/, 3/ Type A @ load of 25 pounds Type B @ load of 20 pounds	 25 25

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TABLE I. Physical and chemical requirements. - Continued

Characteristic	Requirement
Shrinkage temperature at or below 198°F	No shrinkage
Area stability to perspiration, percent (max.) 2/ Subclass A	15
Area stability to laundering, percent (max.) 2/ Subclass A	20
Hexane soluble material, percent (max.) 4/	25.0
Chromic oxide, percent (min.) 4/	2.5
Total ash, percent (max.) 4/	9.0
pH (min.)	3.0

1/ Note: 1 ounce = 1/64 inch.
2/ At least 80 percent of the specimens shall meet the requirement.
3/ Any specimen that ruptures or exhibits grain crack below the specified load shall be recorded as a failing specimen.
4/ Calculated on a moisture-free basis.

3.7.1 Form. The leather shall be in the form of whole skin and shall be trimmed. The area expressed in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather.

3.8 Fungicide (mildew resistance), Class 1 only. Unless otherwise specified (see 6.2), the leather shall be treated with one of the following fungicides to provide mildew resistance. The fungicides to treat the leather shall be an EPA registered product that complies with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (see 2.2.2).

3.8.1 2-(Thiocyanomethylthio) bensothiazole (TCMTB). The leather shall contain not less than 100 ppm and not more than 400 ppm of active ingredient TCMTB when tested as specified in 4.5.

3.8.2 *Ortho*-phenyl phenol (OPP). The leather shall contain not less than 540 ppm and not more than 2580 ppm of active ingredient OPP when tested as specified in 4.5.

3.8.3 Diiodomethyl-*para*-tolylsulfone (DIMPTS). The leather shall contain not less than 300 ppm and not more than 2000 ppm of active ingredient DIMPTS when tested as specified in 4.5.

3.9 Workmanship. The finished leather shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5 and 4.6.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination of the end item. The leather for each type shall be examined and defects shall be classified as stated in Table II. The sample unit for these examinations shall be one (1) sheepskin. The lot size shall be expressed in units of one sheepskin. The contract or purchase order shall identify the number of samples required for thickness for acceptance and rejection of the leather skins. Separate examinations shall be performed for visual and the thickness, but the same sample may be used for each examination.

TABLE II. Visual examination defects.

Examination	Defect 1/
Quality of leather	Any cut, hole, tear, scratch, brand, vein, wrinkle, cockle, coarse loose fiber or flesh side, boardy area, flanky area and other imperfections that individually or collectively cause more than 12 percent loss of the total area of the leather
Color (all types)	Not as specified Not dyed as specified Not uniform
Finish	Unless otherwise specified, grain side not finished. Streaky, flaky, cracked or tacky. Excessively pigmented

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TABLE II. Visual examination defects. - Continued

Examination	Defect 1/
Form	Not whole sheepskin
Area (expressed in terms of square feet)	Not as marked on leather
Marking	Omitted, not readable
Thickness	Not as specified 2/

1/ Light, well healed scratches and grub holes, light fat wrinkles, and slight stains shall not be classified as defects.
2/ Thickness shall be measured in accordance with ASTM D1814.

4.5 End item testing. The finished leather shall be tested for the characteristics as specified in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. The sample grinding and sampling location from which the sample unit is to be obtained shall be in accordance with ASTM D2813. The area of leather from which the pieces are obtained for preparing the test specimens shall be expanded to 12 inches by 12 inches. Fifteen units of product shall be selected at random from each lot for the purpose of testing. For lots consisting of less than 15 units of product, each unit shall be sampled for testing. The sample(s) required for testing the lot for physical and chemical characteristics shall be selected from the samples for visual or dimensional thickness examination. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exists:

a. More than three (3) specimen fail to meet stiffness requirements.
b. More than three (3) specimens fail to meet stitch tearing strength requirements or more than three (3) specimens fail to meet elongation requirements.
c. When Subclass A is specified, more than three (3) specimens fail to meet perspiration resistance requirements.
d. When Subclass A is specified and the leather is tested for area stability to laundering, more than three (3) specimens fail to meet the laundering requirement.
e. One (1) or more specimens fail to meet any of the remaining requirements applicable to the sample unit.
f. Any composite fails to meet the specified requirement.

TABLE III. End item tests.

Characteristic	Requirement reference	Test Method	Requirement	
			Sample unit	Composite sample
Material identification	See 3.3	Visual (see 6.4)	X	-
Thickness	Table I	ASTM D2813	X	-

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TABLE III. End item tests. – Continued

Characteristic	Requirement reference	Test Method	Requirement	
Stiffness	Table I	ASTM D2821	X	-
Colorfastness to crocking	Table I	ASTM D5053 1/	X	-
Staining	Table I	ASTM D6012	X	-
Stitch tear strength	Table I	ASTM D4705	X	-
Elongation	Table I	ASTM D2211	X	-
Shrinkage temperature	Table I	ASTM D6076	X	-
Area stability to perspiration (Subclass A)	Table I	ASTM D2322 2/	X	-
Area stability to laundering (Subclass A)	Table I	ASTM D2096 3/	X	-
Hexane-soluble material	Table I	ASTM D3495	-	X
Chromic oxide	Table I	ASTM D2807	-	X
Total ash	Table I	ASTM D2617	-	X
pH	Table I	ASTM D2810	-	X
Fungicide (Class 1)	See 3.8	ISO 13365-1 4/	-	X

1/ Evaluated using AATCC EP8.
2/ The suction flask shall contain 15 (± 1)mL of artificial perspiration solution per specimen and the circulating air oven shall be preheated to 68 (± 2)ºC.
3/ Only one specimen shall be tested for each 8 by 8 inch sample unit.
4/ When testing DIMPTS, use an HPLC analytical standard to measure the concentration of DIMPTS in accordance with ISO 13365-1.

4.6 Methods of testing and inspection.

4.6.1 Visual shade examination. The color and appearance of the finished leather shall be viewed using AATCC EP9, Option C when viewed under filtered tungsten lamps which approximate artificial daylight D_{75} illuminant having a correlated color temperature of 7500K (± 200 K) with illumination of 100 (± 20) foot- candles, and shall be a good match to the standard sample under incandescent lamplight at 2850 K (± 200 K).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands.

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Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The leather covered by this document is intended primarily for use in dress gloves, flyer's gloves and helmets.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type, class and subclass required (see 1.2)
c. The specific issue of individual documents referenced
d. When first article is required (see 3.1)
e. Color required (see 3.5)
f. When fungicide is required (3.6)
g. Conformance inspection (see 4.3)
h. Inspection conditions (see 4.3.1)
i. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Chrome tanned sheepskin leather may be purchased as a commercial item in the following colors: cream, black, dark brown and saddle brown.

6.6 Subject term (key word) listing.

Domestic
Fungicide
Hair sheep
Mildew resistant

6.7 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CIVIL AGENCY COORDINATING ACTIVITY
GSA-FAS

Custodians:
Army - GL
Navy– NU
Air Force – 11

Preparing Activity:
DLA-CT

(Project: 8330-2022-001)

Review activities:
Army – MD

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